

# Math 421

Friday, October 24

## Announcements

- Homework 6 Due **today**

# lim sup and lim inf

$$S_N = \{s_n : n > N\}$$

$$S_1 \supset S_2 \supset S_3 \dots$$

$$u_1 \leq u_2 \leq u_3 \leq \dots$$

$(u_N)$   
increasing

$$v_1 \geq v_2 \geq v_3 \geq \dots$$

**Def.** Let  $(s_n)$  be a sequence in  $\mathbb{R}$ . We define

$$\liminf s_n = u = \lim_{N \rightarrow \infty} u_N = \lim_{N \rightarrow \infty} \inf\{s_n : n > N\}$$

and

notation

$$\limsup s_n = v = \lim_{N \rightarrow \infty} v_N = \lim_{N \rightarrow \infty} \sup\{s_n : n > N\}$$

$(v_N)$  decreasing sequence.

**Examples.**  $(s_n)$   $s_n = \frac{1}{n}$   $\liminf s_n = 0$   $\limsup s_n = 0$ .

$$(s_n) = (-1, 0, 1, 0, -1, \dots)$$

## Activity

Let  $(s_n)$  be the sequence defined by  $s_n = \sin\left(\frac{3\pi n}{2}\right)$ .

1. Determine  $S_N = \{s_n : n > N\}$  for some explicit values of  $N$ .

$$S_1 = \{0, 1, 0, -1, \dots\} \quad S_4 = \{-1, 0, 1, 0, -1, \dots\}$$

2. Determine  $u_N = \inf\{s_n : n > N\}$  and  $v_N = \sup\{s_n : n > N\}$ .

$$u_N = -1 \quad \text{for all } N \quad v_N = 1 \quad (u_N) = (-1, -1, -1, -1, \dots) \\ (v_N) = (1, 1, 1, 1, \dots)$$

3. Compute  $\liminf s_n$  and  $\limsup s_n$ .

$$\liminf s_n = -1 \in A \quad \liminf s_n = \inf A$$

$$\limsup s_n = 1 \in A \quad \limsup s_n = \sup A$$

4. How are  $\liminf s_n$  and  $\limsup s_n$  related to the set of subsequential limits?

$$A = \{-1, 0, 1\}$$

$$s_n = \frac{1}{n} \quad \limsup s_n < \sup \{s_n : n \in \mathbb{N}\}$$

$$t_n = \frac{3\pi n}{2} \quad \limsup t_n = \sup \{t_n : n \in \mathbb{N}\}$$

## Discussion Questions

Convince yourselves the following properties are true:

❖  $\limsup s_n \leq \sup \{s_n : n \in \mathbb{N}\} \text{ \& \; } \liminf s_n \geq \inf \{s_n : n \in \mathbb{N}\}$

❖  $\liminf s_n \leq \limsup s_n$

$$u_N \leq v_N$$

❖ If  $\lim s_n$  exists then

❖  $u_N \leq \lim s_n \leq v_N$  for all  $N$

❖  $\liminf s_n \leq \lim s_n \leq \limsup s_n$

$$\lim s_n = \liminf s_n = \limsup s_n$$

**Theorem 10.7:** Let  $(s_n)$  be a sequence in  $\mathbb{R}$ ,  $(s_n)$  has a limit if and only if  $\liminf s_n = \limsup s_n$ .

*Discussion.*

$(\Rightarrow)$  If  $\lim s_n$  exists then  $\liminf s_n = \limsup s_n = \lim s_n$   
 • Infinite limits

• Finite limits

$(\Leftarrow)$  If  $\liminf s_n = \limsup s_n$  then  $\lim s_n$  exists and

$\lim s_n = \liminf s_n = \limsup s_n$ .

• Infinite limits

• Finite limits.

$$\lim s_n = \liminf s_n = \limsup s_n$$

**Theorem 10.7:** Let  $(s_n)$  be a sequence in  $\mathbb{R}$ ,  $(s_n)$  has a limit if and only if  $\liminf s_n = \limsup s_n$ .

**Proof ( $\Rightarrow$ ).** Finite limits.  $\lim s_n = s \Rightarrow \liminf s_n = s = \limsup s_n$ .

Let  $\varepsilon > 0$ . Since  $\lim s_n = s$ , there exists  $N \in \mathbb{N}$  such that  $n > N$

$$|s_n - s| < \varepsilon. \text{ So } -\varepsilon < s_n - s < \varepsilon \text{ and } s - \varepsilon < s_n < s + \varepsilon$$

$$v_N = \sup \{s_n : n > N\} \leq s + \varepsilon. \text{ Thus for } m > N \quad v_m \leq s + \varepsilon$$

and so  $\lim v_m \leq s + \varepsilon$ , as proven in HW Q6, and

$$\limsup s_n \leq s + \varepsilon \text{ for all } \varepsilon > 0. \text{ Thus } \underline{\limsup s_n \leq s}.$$

$$\begin{aligned} \limsup s_n &\leq \lim s_n \\ \text{Similar argument.} \\ \liminf s_n &\geq \lim s_n. \end{aligned}$$

$$\left[ \begin{array}{l} \text{if not there exists } \varepsilon_0 \text{ such} \\ \text{that } \limsup s_n - s > \varepsilon_0 \\ \Rightarrow \limsup s_n > s + \varepsilon_0 \end{array} \right.$$